can be reproduced, so that its outcome depends neither on the personality of the individual performing the experiment, nor on the place and time of the measurement, nor on any other attendant circumstance. But this simply means that the factor which is decisive for the result of the measurement lies beyond the observer, and that one is therefore necessarily led to questions concerning real causal connections operating independently of the observer.

To be sure, it must be agreed that the positivistic outlook possesses a distinctive value; for it is instrumental to a conceptual clarification of the significance of physical laws, to a separation of that which is empirically proven from that which is not, to an elimination of emotional prejudices nurtured solely by customary views, and it thus helps to clear the road for the onward drive of research. But Positivism lacks the driving force for serving as a leader on this road. True, it is able to eliminate obstacles, but it cannot turn them into productive factors. For its activity is essentially critical, its glance is directed backward. But progress, advancement requires new associations of ideas and new queries, not based on the results of measurements alone, but going beyond them, and toward such things the fundamental attitude of Positivism is one of aloofness.

Therefore, up to quite recently, positivists of all hues have also put up the strongest resistance to the introduction of atomic hypotheses and thereby also to the acceptance of the above mentioned universal constants. This is quite understandable, for the existence of these constants is a palpable proof of the existence in nature of something real and independent of every human measurement.

Of course, even to-day a consistent positivist could call the universal constants mere inventions which have proved to be uncommonly useful in making possible an accurate and complete description of the most diversified results of measurements. But hardly any real physicist would take such an assertion seriously. The universal constants were not invented for reasons of practical convenience, but have forced themselves upon us irresistibly because of the agreement between the results of all relevant measurements, and—this is the essential thing—we know quite well in advance that all future measurements will lead to these selfsame constants.

To sum it all up, we can say that physical science demands that we admit the existence of a real world independent from us, a world which we can however never recognize directly but can apprehend only through the medium of our sense experiences and of the measurements mediated by them.

If we pursue this principle further, our outlook on the world takes a different form. The subject of the observation, the observing Ego, loses its position at the focus of thought and is relegated to a quite modest place. In fact, how pitifully small, how powerless we human beings must appear to ourselves if we stop to think that the planet Earth on which we live our lives